

GHANITHAN SUBRAMANI

PRINCIPAL SOFTWARE ENGINEER

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PROFESSIONAL SUMMARY: Senior Technologist with **15+ years** of expertise in high-performance backend development, cloud-native architectures, and microservices. Specialized in **Rust, Golang, platform security and performance engineering** with proven track record of optimizing enterprise systems for scale and reliability. **Google Professional Cloud Architect** driving technical innovation across fintech, media, and enterprise platforms.

KEY QUALIFICATIONS

- **Architecture & Performance:** Designed high-performance systems handling massive transaction volumes with significant deployment efficiency improvements using **Rust, Golang, and cloud-native technologies**
- **DevOps & Security:** Implemented comprehensive **CI/CD pipelines, security toolchains (CSPM, OWASP ZAP)**, and **multi-cloud observability** with OpenTelemetry and Prometheus
- **Technical Leadership:** Led cross-functional teams, established best practices, and mentored engineers across **startup to enterprise environments**
- **Full-Stack Expertise:** End-to-end development from **microservices architecture** to **Progressive Web Apps**, covering diverse technology stacks and industry domains

PROFESSIONAL EXPERIENCE

Principal Software Engineer OpsMx Inc.

Nov 2024 - Present

Key Achievements:

- Architected and developed virtual application packaging solution using **Packer, K3s, and bash scripting**, reducing deployment time from 8 hours to 5 minutes - a **99% improvement** in installation efficiency
- Created process for cross-platform **OVA distribution** package enabling seamless deployment across **GCP, AWS, and VMware** environments, streamlining customer onboarding processes
- Designed and implemented a global service for collecting and rendering OSS library metrics and vulnerabilities, providing centralized security insights for all customer deployments
- Enhanced **ZAP security scanning** capabilities to support API testing using **OpenAPI specifications v2 and v3**, expanding automated security testing coverage
- Increased platform observability using **OpenTelemetry with collector for aggregation and Prometheus**, improving system monitoring and operational insights
- Owned architecture across **authentication and authorisation, audit, and cloud security posture**, covering the token model, session handling and the authorisation model
- Designed the audit service and delivered it to other services as a **Go package with a fluent API**, reducing event instrumentation to a single expression at the call site
- Built a context service exposing cloud security posture to **AI agents over MCP**, with its own query language and a TTL-based lifecycle

CORE COMPETENCIES

Software Architecture & Development

Requirements Analysis

Solution Design

Team Leadership & Management

UI/UX Design

DevOps Automation & Tooling

Multi-Cloud Distribution

CERTIFICATIONS

Google Professional Cloud Architect

Google Cloud Platform
[View Certificate](#)
2025

Google Cloud Essentials

Build Infrastructure with Terraform on Google Cloud

Fellow, Insurance Institute of India (FE93778)

Distinction in Actuary

EDUCATION

- Led [Dgraph](#) performance engineering: reduced pod restarts, sized memory, block cache and index allocation against production workloads, and drove query latency toward [1-2 seconds](#)
- Built a customer-deployable diagnostic toolset that collects database and node telemetry into a single report with [sizing recommendations](#), with an accompanying investigation runbook
- Delivered [GitHub App integration](#) for on-premises installations, covering credential distribution and key rotation
- Established the architecture documentation and delivery plan, since adopted and owned by the wider engineering team
- Integrated comprehensive security toolchain including [ScoutSuite \(CSPM\)](#) and network reachability with the platform across multi-cloud environments

• Principal Software Engineer

Astra Technologies Inc.

Jun 2023 - Sep 2024

Key Achievements:

- Built a [low latency](#) market data aggregation service and Over the Counter (OTC) trading solution with a very small team in the early stage start-up
- Architected and developed a high-performance, low-latency backend (REST and WebSocket) using Idiomatic [Rust](#) and async libraries, including [Tokio](#), [Tungstenite](#), [Axum](#)
- Set up monitoring and traceability systems with [OpenTelemetry](#), [Jaeger](#), [Prometheus](#), [Grafana](#) and [Honeycomb](#)
- Worked with [Docker](#) and cross compiled binaries to deploy services in [AWS cloud](#)
- Led DevOps team in designing the [AWS VPC](#) architecture, ensuring a secure, scalable and cost-effective infrastructure
- Developed CI/CD pipelines using [GitHub workflows](#)
- Built web applications using [Lit](#), [Web Components](#), [React](#), [Next.js](#), and [Firebase](#)
- Created [Bitcoin Wallet Service](#) from the ground up in Rust
- Worked in [agile environment with sprint planning](#) and iterative development cycles typical of early-stage fintech startup
- Led architectural design process to ensure scalable, efficient, and maintainable solutions

• Senior Software Engineer

XRG Consulting Pvt. Ltd. - Quickplay Media

Sep 2022 - Jun 2023

Key Achievements:

- Developed [Golang microservices](#) for Content Management System serving OTT platform provider, Quickplay Media (client: [Allen Media Group](#))
- Worked extensively with [Kubernetes](#) and used [GCP](#) as cloud provider
- Optimized backend to handle [60,000 TPS](#) during critical events such as Hurricanes in US
- Developed sidecar solution for image processing, reducing cost by [50%](#)
- Collaborated with cross-functional teams including product management, QA and operations to align technical solutions with business goals

• Risk Underwriter / Lead Fullstack Engineer

United India Insurance Company Ltd.

Nov 2016 - Sep 2022

Key Achievements:

- Developed internal Document Management System using [Go \(Gin\)](#) and [MongoDB](#)
- Created [offline-first Progressive Web App](#) for premium calculator using [Svelte](#), [Vite](#), [PouchDB](#), [GO \(Fiber\)](#), [CouchDB](#)
- Led cross-functional teams for Digilocker integration and claims reporting for banks
- Created phonetic name comparison algorithm (REST API) for Indian names using Soundex-like approach

• Senior Software Engineer

Bosch Global Software Technologies

Sep 2014 - Sep 2016

Key Achievements:

- Implemented [CAN injection](#) for Diesel Truck ECU hardware encapsulation layer

• Sr. Design and Development Engineer

HandsOn Technologies

Apr 2013 - Aug 2014

Bachelor of Engineering

Electronics and

Communication

Kumaraguru College of

Technology

2011

TECHNICAL EXPERTISE

Programming Languages:

Rust, Go, JavaScript, C, Python, Java, C++

Database Systems:

PostgreSQL, Couchbase, Oracle, MySQL, MongoDB, Dgraph

Web Technologies:

REST/WebSocket APIs, JWT, PWA, Kubernetes, GraphQL, gRPC, OpenTelemetry

Frameworks:

Tokio, Axum, Actix, Next.js, Lit, Svelte, Express.js, Fiber

DevOps & Tools:

Prometheus, Jaeger, Grafana, Docker, Packer, K3s, GitHub Workflows, Terraform

Security & Compliance:

ScoutSuite (CSPM), ZAP (OWASP), OpenAPI Security Testing, Vulnerability Management

Cloud Platforms:

Google Cloud Platform (PCA), AWS (Complete VPC Setup)

Methodologies:

Agile, Sprint Planning, DevOps, Microservices Architecture

NOTABLE PROJECTS

- ▶ [Rust Tour](#) - interactive Rust course, 40 test-driven exercises, built in Rust
- ▶ [archMD](#) - single-file browser Markdown editor with Mermaid and PDF export
- ▶ [Interactive console MP3 player](#) in C++
- ▶ [PDFReplicator](#) - Million PDF generation system using Java
- ▶ [Intelligent Traffic Control System](#) using ZigBee Mesh technology

Key Achievements:

- Developed communication protocol libraries (CAN, I2C, SPI, UART) and ARM Cortex-M0 firmware
- Designed 6 major boards , plus two products taken from requirement through to production in weight-based industrial automation and home automation

Software Engineer

Bosch Global Software Technologies

Jun 2011 - Mar 2013

Key Achievements:

- Developed Device Encapsulation Library with temperature/pressure sensors using PERL scripts and created device drivers

PERSONAL INFORMATION

Location: Chennai, India
Languages: English, Tamil